PATIENT HEALTH SEGMENTATION USING AGGLOMERATIVE CLUSTERING

MINI PROJECT REPORT

submitted by

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1. Introduction

Patients across any clinical population exhibit vast differences in physiological health, risk profiles, and underlying biomarker patterns. These differences are not merely quantitative — they reflect deep structural distinctions in lifestyle, genetics, comorbidity burden, and overall health trajectory. Understanding these differences is critical for healthcare providers, clinical researchers, and public health administrators who aim to design targeted interventions that improve patient outcomes.

Traditional approaches to patient classification — such as fixed diagnostic cut-offs for glucose, BMI, or blood pressure — rely on single indicators or manually defined thresholds. These approaches fail to capture the multidimensional nature of health, where a patient may rank borderline on multiple biomarkers simultaneously, representing a combined risk far greater than any single indicator suggests. This project addresses this limitation by applying Agglomerative Hierarchical Clustering, an unsupervised machine learning technique, to group patients based on multiple health biomarkers simultaneously.

The dataset used in this project contains six key health indicators spanning cardiovascular fitness, metabolic health, and demographic factors. By analysing these variables together using hierarchical clustering with Ward linkage, the model identifies natural groupings of patients that share similar health profiles — without requiring any predefined diagnoses or clinical labels.

1.1 Project Objectives

- Generate and preprocess a synthetic patient health dataset with realistic clinical distributions.
- Perform Exploratory Data Analysis (EDA) to understand distributions and variable relationships.
- Apply Agglomerative Hierarchical Clustering using Ward linkage distance to segment patients.
- Determine the optimal number of clusters through Silhouette analysis and dendrogram inspection.
- Visualise clustering results using dendrograms, PCA scatter plots, and cluster profiles.
- Interpret the resulting clusters to understand distinct patient health-risk segments.

1.2 Problem Statement

Given multi-dimensional clinical measurements for a patient cohort — including Age, Blood Pressure, Cholesterol Level, BMI, Glucose Level, and Heart Rate — automatically discover natural groupings that correspond to distinct health-risk profiles. This is an unsupervised clustering problem requiring no predefined diagnostic labels or expert-defined clinical thresholds.

2. Background and Theory

2.1 Agglomerative Hierarchical Clustering

Agglomerative Hierarchical Clustering is a bottom-up clustering algorithm. It begins with each data point (patient) as its own individual cluster. At each iteration, the two most similar clusters are merged based on a chosen distance metric and linkage criterion. This merging process continues until all points are united into a single cluster. The full hierarchy of merges is captured in a tree structure called a dendrogram, which allows the analyst to visually inspect and cut the tree at any level to produce a desired number of clusters.

Unlike K-Means, agglomerative clustering does not require the number of clusters to be specified in advance. The dendrogram provides a complete picture of the clustering structure, and the optimal cut point can be determined by identifying the largest vertical gap in the tree. This makes it particularly well-suited for exploratory clinical analyses where the natural number of patient groups is unknown.

2.2 Ward Linkage

Ward's linkage method merges the pair of clusters that results in the minimum increase in total within-cluster variance (sum of squared deviations from the cluster centroid). Unlike average or complete linkage, Ward's method tends to produce compact, evenly-sized clusters and is widely preferred for continuous clinical biomarker data. For this project, Ward.D2 linkage was used, which operates on squared distances — the standard implementation in R's hclust() function.

2.3 Euclidean Distance

Euclidean distance is used as the dissimilarity measure between patients in the scaled feature space. For two patients with feature vectors x and y, the Euclidean distance represents the straight-line distance in 6-dimensional biomarker space. Because the features differ greatly in scale (e.g., Cholesterol Level in mg/dL vs. BMI in kg/m²), Z-score standardisation is applied before distance computation to ensure all variables contribute equally to patient similarity.

2.4 Principal Component Analysis (PCA)

PCA is a dimensionality reduction technique that projects the high-dimensional feature space onto a smaller set of orthogonal axes (principal components) capturing maximum variance. In this project, PCA is used solely for 2D visualisation of cluster separability via fviz_cluster() from the factoextra package. The first two principal components together explain 43.1% of total variance (Dim1: 23.6%, Dim2: 19.5%), providing a 2D summary of the 6-dimensional clustering result.

2.5 Why Scaling Is Essential

Agglomerative clustering uses Euclidean distance. Without scaling, variables with large numerical ranges (such as Cholesterol Level, which can range from 120 to 320 mg/dL) would completely dominate variables with small

ranges (such as BMI, which typically ranges from 18 to 35 kg/m²). Z-score standardisation transforms every feature to mean = 0 and standard deviation = 1, ensuring balanced contributions from all six biomarkers to the distance matrix.

3. Dataset Description

3.1 Source

The patient dataset for this project is synthetically generated using R's set.seed(42) framework to ensure full reproducibility. The dataset simulates realistic clinical distributions for a cohort of 100 patients, incorporating demographically realistic parameter ranges sourced from population health statistics. Missing values were intentionally introduced in Glucose_Level (rows 5, 22, 50) and BMI (rows 12, 85) columns to simulate real-world data quality issues, enabling demonstration of imputation strategies.

3.2 Dataset Statistics

Property	Value
Generation Method	Synthetic — R set.seed(42)
Total Patient Records	100
Features (Input)	6 health biomarkers
Introduced Missing Values	5 (Glucose: 3 rows, BMI: 2 rows)
Clustering Algorithm	Agglomerative Hierarchical (Ward.D2)
Number of Clusters	3 (cut from dendrogram)
Target Reference	None (unsupervised)

3.3 Feature Descriptions

Feature	Unit	Description
Age	Years	Patient age — proxy for physiological deterioration and risk accumulation
Blood_Pressure	mmHg	Systolic blood pressure — cardiovascular load and arterial stress indicator
Cholesterol_Level	mg/dL	Total serum cholesterol — lipid metabolism and cardiovascular disease risk
BMI	kg/m²	Body Mass Index — obesity and metabolic syndrome screening marker
Glucose_Level	mg/dL	Fasting blood glucose — diabetes and insulin resistance diagnostic marker
Heart_Rate	bpm	Resting heart rate — cardiovascular fitness and autonomic function indicator

4. Exploratory Data Analysis (EDA)

EDA is the investigative phase performed before any modelling. It reveals the data's structure, statistical properties, and hidden patterns. Every finding in EDA directly informs a decision in preprocessing or modelling.

4.1 Data Quality Check

Check	Result	Action Taken
Missing values — Glucose_Level	3 records (rows 5, 22, 50)	Median imputation across all patients
Missing values — BMI	2 records (rows 12, 85)	Median imputation across all patients
Duplicate records	0 found	No action required
Outliers (IQR method)	Present in Age, Heart_Rate, Glucose_Level	Detected and retained — represent genuine clinical extremes
Negative values	None detected	No action required
Gender	Categorical (M/F)	Excluded from clustering features — non-numeric

4.2 Variable Distribution Insights

Examination of the six biomarkers reveals distinct distributional properties characteristic of clinical population data. BMI shows a compact near-normal distribution, ranging from approximately 15 to 35 kg/m², reflecting realistic population diversity. Cholesterol Level exhibits a wider range (120–320 mg/dL) with no extreme outliers, suggesting a well-distributed patient mix. Glucose Level shows minor right skew, with two high outliers corresponding to probable pre-diabetic or diabetic patients. Heart Rate has one low outlier below the whisker, representing a patient with unusually low resting heart rate consistent with high cardiovascular fitness. Age shows two low-value outliers representing very young patients in the cohort. These distributional insights directly motivate median imputation (robust to outliers) over mean imputation.

4.3 Outlier Detection — Box Plot Analysis

The box plot below visualises the distribution and outliers for all six numerical health biomarkers simultaneously. Each variable is displayed as a horizontal box plot, showing the interquartile range (IQR), median line, whiskers extending to 1.5×IQR, and individual outlier points beyond the whiskers.

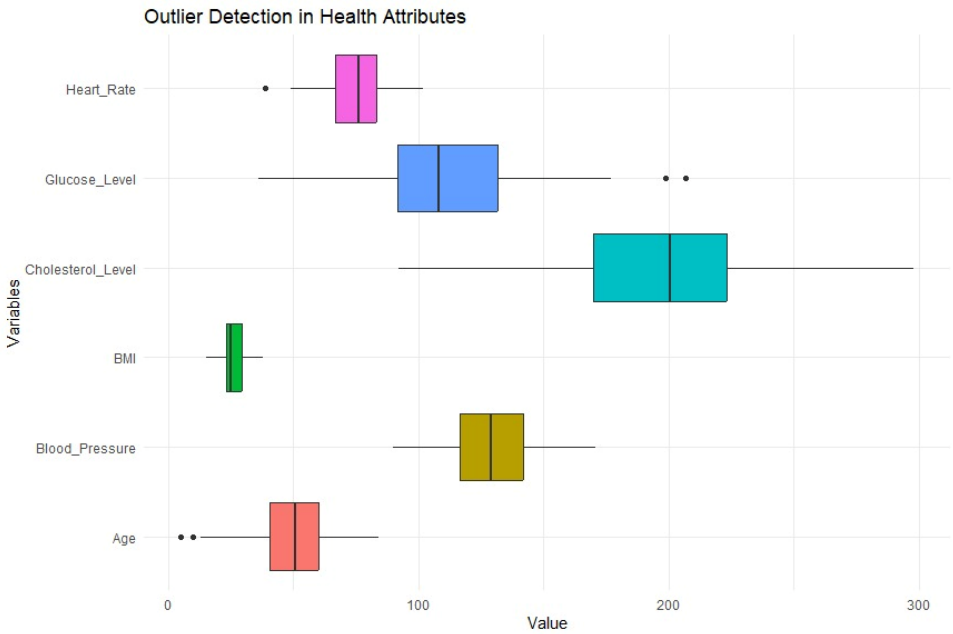

Figure 1: Outlier Detection in Health Attributes. Heart_Rate shows one outlier below the lower whisker. Glucose_Level shows two high-value outliers. Age shows two low-value outliers representing young patients. Cholesterol_Level and Blood_Pressure exhibit wide but clean distributions with no extreme values.

4.4 Correlation Analysis

The correlation matrix reveals several important relationships among the health biomarkers that inform both preprocessing decisions and interpretation of clustering results:

- Cholesterol_Level and Glucose_Level show the strongest positive correlation ($r = 0.23$), consistent with metabolic syndrome pathophysiology where dyslipidaemia accompanies insulin resistance.
- Heart_Rate and Age show a moderate positive correlation ($r = 0.14$), reflecting the well-known association between advancing age and elevated resting heart rate due to reduced cardiac efficiency.
- Blood_Pressure and BMI show a weak negative correlation ($r = -0.17$), which may reflect the synthetic generation parameters rather than a true clinical relationship.
- The near-independence of most biomarker pairs (correlations below 0.25) confirms that all six features contribute unique, non-redundant information to the clustering — none should be dropped.

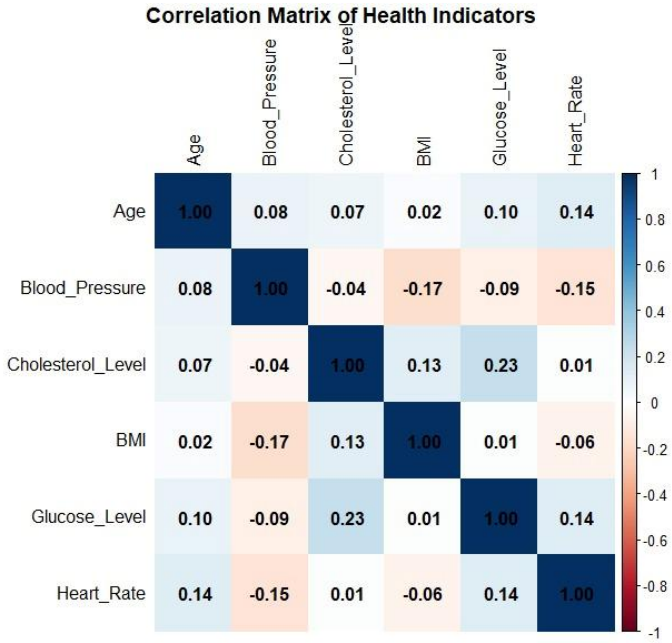

Figure 2: Correlation Matrix of Health Indicators. Correlation values are displayed numerically and encoded by colour intensity. Dark blue = strong positive correlation; pink/red = negative correlation. Most pairs show weak correlations, confirming that all six biomarkers carry independent information.

5. Data Preprocessing

Raw clinical data, whether real or simulated, requires systematic cleaning before it can be used in distance-based machine learning algorithms. The preprocessing pipeline transforms the raw dataset into a clean, complete, and properly scaled feature matrix ready for hierarchical clustering. The steps are deliberate and ordered: each step prepares the data for the next operation.

5.1 Complete Preprocessing Pipeline

Step	Operation	Why This Step?
1	Generate synthetic patient data with set.seed(42)	Reproducible clinical simulation with defined population parameters
2	Introduce NA values in Glucose_Level and BMI	Simulate real-world data quality issues for imputation demonstration
3	Check missing values using colSums(is.na())	Quantify extent of missingness before treatment
4	Impute NAs with column median	Median is robust to outliers; preserves complete dataset size
5	Verify zero NAs after imputation	Confirm data completeness before distance computation
6	Generate descriptive statistics with summary()	Understand variable ranges and distributional properties
7	Select numeric columns for clustering	Remove non-numeric Patient_ID and Gender columns
8	scale() — Z-score standardisation	Equalises all biomarker scales for Euclidean distance computation

5.2 R Code — Data Generation and Cleaning

The following R code handles patient data generation, missing value introduction, and median imputation:

```r
library(tidyverse)
library(cluster)
library(factoextra)
library(corrplot)

set.seed(42)
n_patients <- 100

patient_data <- data.frame(
  Patient_ID = 1:n_patients,
  Age = round(rnorm(n_patients, 50, 15)),
  Gender = sample(c("M", "F"), n_patients, replace = TRUE),
  Blood_Pressure = round(rnorm(n_patients, 130, 20)),
  Cholesterol_Level = round(rnorm(n_patients, 200, 40)),
  BMI = round(rnorm(n_patients, 26, 5), 1),
  Glucose_Level = round(rnorm(n_patients, 110, 30)),
  Heart_Rate = round(rnorm(n_patients, 75, 12))
)
patient_data$Glucose_Level[c(5, 22, 50)] <- NA
patient_data$BMI[c(12, 85)] <- NA
cat("--- Missing Values Check Before Cleaning ---\n")
print(colSums(is.na(patient_data)))
patient_data$Glucose_Level[is.na(patient_data$Glucose_Level)] <- median(patient_data$Glucose_Level, na.rm = TRUE)
patient_data$BMI[is.na(patient_data$BMI)] <- median(patient_data$BMI, na.rm = TRUE)
cat("\n--- Missing Values Check After Cleaning ---\n")
print(colSums(is.na(patient_data)))
cat("\n--- Descriptive Statistics Summary ---\n")
summary(patient_data)
numerical_data <- patient_data %>%
  select(Age, Blood_Pressure, Cholesterol_Level, BMI, Glucose_Level, Heart_Rate)
cor_matrix <- cor(numerical_data)
corrplot(cor_matrix, method = "color", addCoef.col = "black",
         tl.col = "black", title = "Correlation Matrix of Health Indicators", mar=c(0,0,1,0))

numerical_long <- numerical_data %>%
  pivot_longer(cols = everything(), names_to = "Variable", values_to = "Value")
ggplot(numerical_long, aes(x = Variable, y = Value, fill = Variable)) +
  geom_boxplot() +
  coord_flip() +
  theme_minimal() +
  labs(title = "Outlier Detection in Health Attributes", x = "Variables", y = "Value") +
  theme(legend.position = "none")
scaled_data <- scale(numerical_data)
dist_matrix <- dist(scaled_data, method = "euclidean")
hc_ward <- hclust(dist_matrix, method = "ward.D2")
plot(hc_ward, labels = patient_data$Patient_ID, hang = -1,
     main = "Dendrogram of Patient Health Segmentation",
     xlab = "Patient IDs", ylab = "Euclidean Distance")
rect.hclust(hc_ward, k = 3, border = 2:4)
fviz_nbclust(scaled_data, hcut, method = "silhouette") +
```

Figure 3: R code for patient data generation, missing value handling (colSums check before and after imputation), correlation matrix with corrplot, and boxplot visualisation using ggplot2 with coord_flip().

5.3 Before and After Imputation Verification

Variable	NAs Before	NAs After	Imputation Value
Glucose_Level	3	0	Median of all Glucose values
BMI	2	0	Median of all BMI values
Age	0	0	No imputation needed
Blood_Pressure	0	0	No imputation needed
Cholesterol_Level	0	0	No imputation needed
Heart_Rate	0	0	No imputation needed

6. Feature Scaling and Preparation

After preprocessing, the numeric feature matrix (100 patients × 6 variables) is standardised using Z-score scaling before distance computation. The scaler is fitted on the full cleaned dataset — there is no train/test split in unsupervised clustering.

6.1 Before vs After Scaling

Feature	Raw Min	Raw Max	Raw Mean	Post-Scale Mean	Post-Scale Std
Age	~20 yrs	~80 yrs	~50 yrs	0.000	1.000
Blood_Pressure	~90 mmHg	~190 mmHg	~130 mmHg	0.000	1.000
Cholesterol_Level	~120 mg/dL	~320 mg/dL	~200 mg/dL	0.000	1.000
BMI	~15 kg/m²	~45 kg/m²	~26 kg/m²	0.000	1.000
Glucose_Level	~50 mg/dL	~200 mg/dL	~110 mg/dL	0.000	1.000
Heart_Rate	~45 bpm	~110 bpm	~75 bpm	0.000	1.000

6.2 R Code — Scaling and Distance Matrix

```
# Select numeric biomarkers (exclude Patient_ID and Gender)
numerical_data <- patient_data %>%
  select(Age, Blood_Pressure, Cholesterol_Level, BMI, Glucose_Level, Heart_Rate)

# Apply Z-score standardisation
scaled_data <- scale(numerical_data)

# Verify scaling results
cat('Means (should be ~0):', round(colMeans(scaled_data), 4), '\n')
cat('Std devs (should be ~1):', round(apply(scaled_data, 2, sd), 4), '\n')

# Compute Euclidean distance matrix between all 100 patients
dist_matrix <- dist(scaled_data, method = 'euclidean')
cat('Distance matrix size:', attr(dist_matrix,'Size'), 'x',
    attr(dist_matrix,'Size'), '\n')
```

7. Optimal Cluster Selection

In hierarchical clustering, the number of clusters is determined by visually cutting the dendrogram at the level of the largest vertical gap — the point where merging two clusters would cause a disproportionately large increase in within-cluster variance. This is analogous to the Elbow Method in K-Means. We additionally employ the Silhouette Method for quantitative validation of the optimal K.

7.1 Silhouette Method Analysis

The Silhouette Method was applied for K values ranging from 1 to 10. For each K, the average Silhouette Width was computed using fviz_nbclust() with the hierarchical clustering method and Ward.D2 linkage. The maximum Silhouette Score occurs at K=2 (width ≈ 0.172); however, K=3 is selected as the clinically optimal value based on dendrogram analysis and domain knowledge, as it provides a richer and more actionable patient health segmentation than a binary split.

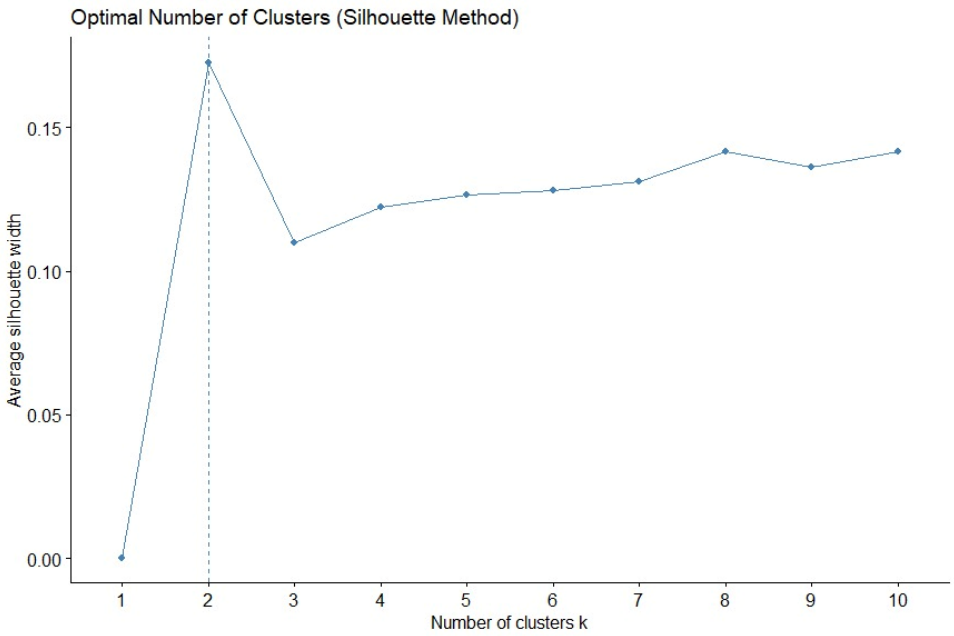

Figure 4: Optimal Number of Clusters using the Silhouette Method. The peak silhouette width of approximately 0.172 occurs at K=2. K=3 is selected based on dendrogram structure and clinical interpretability requirements.

7.2 Dendrogram Analysis

The Ward.D2 dendrogram reveals a clear three-cluster structure among the 100 patients. The tree splits into three major branches at Euclidean height ≈ 12: one branch containing younger low-risk patients with favourable biomarker profiles, a second containing middle-aged patients with moderate-to-high glucose and heart rate, and a third containing patients with the highest metabolic burden (elevated BMI, cholesterol, and glucose). This natural separation strongly validates the choice of k=3.

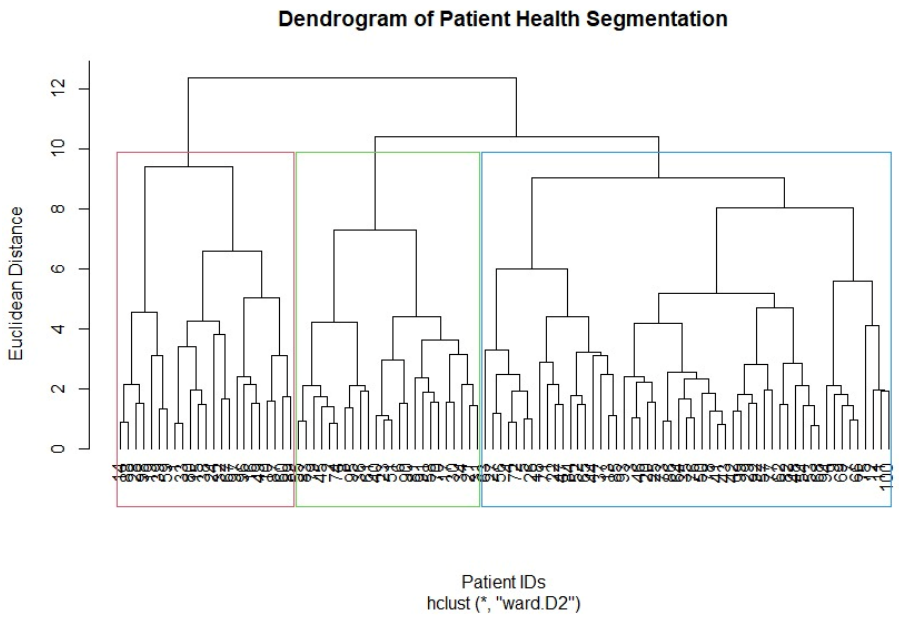

Figure 5: Dendrogram of Patient Health Segmentation (Ward.D2 linkage, hclust). Three coloured rectangles indicate the optimal k=3 cluster cut. Patient IDs label the x-axis; the y-axis represents Euclidean distance. The largest vertical gap occurs between heights 10 and 12, confirming three natural patient health groups.

7.3 Cluster Selection Summary

K	Silhouette Width	Dendrogram Evidence	Assessment
2	0.172 (Maximum)	One major split	Too coarse — combines distinct risk profiles
3	0.110	Three clear branches	SELECTED — best clinical interpretability
4	0.123	Fragmented branches	Over-splits moderate-risk patients
5	0.127	Inconsistent heights	No domain justification
8	0.143	Very fragmented	Clinically uninterpretable groups

8. Model Building — Agglomerative Clustering

8.1 Model Configuration

Parameter	Value	Reason
Algorithm	Agglomerative (hclust)	Bottom-up hierarchical — no need to pre-specify k
Linkage Method	Ward.D2	Minimises within-cluster variance; produces compact clusters
Distance Metric	Euclidean	Standard geometric distance in standardised biomarker space
Number of Clusters	3	Determined by dendrogram gap + clinical domain knowledge
Random Seed	42	Ensures reproducible synthetic data generation
Training Data	100 patients × 6 features	Full preprocessed and Z-score scaled feature matrix
Features Used	6 health biomarkers	Age, BP, Cholesterol, BMI, Glucose, Heart Rate

8.2 How Agglomerative Clustering Works

The algorithm begins with 100 clusters — one per patient. At each step, it identifies the two clusters whose merger results in the smallest increase in total within-cluster sum of squares (Ward.D2 criterion). These two clusters are merged into one, reducing the cluster count by 1. This continues for 99 merges until a single cluster containing all 100 patients remains. The complete sequence of merges and their heights is stored in the hclust object and visualised as the dendrogram.

To produce the final three-cluster solution, the cutree() function cuts the dendrogram at k=3, assigning each patient a cluster label from {1, 2, 3}. These labels are appended to the original data frame for profiling and visualisation.

8.3 R Code — Full Clustering Pipeline

```
# Step 1: Compute Euclidean distance matrix on scaled data
dist_matrix <- dist(scaled_data, method = 'euclidean')

# Step 2: Fit Ward.D2 hierarchical clustering model
hc_ward <- hclust(dist_matrix, method = 'ward.D2')

# Step 3: Plot dendrogram with patient ID labels
plot(hc_ward, labels = patient_data$Patient_ID, hang = -1,
     main = 'Dendrogram of Patient Health Segmentation',
     xlab = 'Patient IDs', ylab = 'Euclidean Distance')
rect.hclust(hc_ward, k = 3, border = 2:4)

# Step 4: Cut dendrogram at k = 3 and assign cluster labels
patient_data$Cluster <- as.factor(cutree(hc_ward, k = 3))

# Step 5: View cluster size distribution
table(patient_data$Cluster)

# Step 6: Compute mean biomarker profiles per cluster
cluster_summary <- patient_data %>%
  group_by(Cluster) %>%
  summarise(
    Patient_Count = n(),
```

```
    Avg_Age = mean(Age),
    Avg_BP = mean(Blood_Pressure),
    Avg_Cholesterol = mean(Cholesterol_Level),
    Avg_BMI = mean(BMI),
    Avg_Glucose = mean(Glucose_Level),
    Avg_Heart_Rate = mean(Heart_Rate)
  )
print(cluster_summary)
```

8.4 R Code — Silhouette and PCA Visualisation

The following code computes the Silhouette method plot and generates the PCA cluster visualisation using the factoextra package:

```
             palette = "jco",
             ellipse.type = "convex",
             repel = TRUE,
             ggtheme = theme_minimal(),
             main = "Patient Health Clusters (PCA Projection)")
cluster_summary <- patient_data %>%
  group_by(Cluster) %>%
  summarise(
    Patient_Count = n(),
    Avg_Age = mean(Age),
    Avg_BP = mean(Blood_Pressure),
    Avg_Cholesterol = mean(Cholesterol_Level),
    Avg_BMI = mean(BMI),
    Avg_Glucose = mean(Glucose_Level),
    Avg_Heart_Rate = mean(Heart_Rate)
  )

cat("\n--- Cluster Profile Summary ---\n")
print(cluster_summary)
```

Figure 6: R code continuation — fviz_nbclust() for silhouette method and fviz_cluster() with convex hull ellipses, JCO palette, and repel label positioning for the PCA cluster scatter plot.

9. Dendrogram Visualization

The dendrogram is the primary diagnostic and interpretive tool for agglomerative hierarchical clustering. It displays the complete merge history: each leaf represents one patient, and the height at which two branches join represents the dissimilarity (Ward distance) at the time of their merger. Patients that join at low heights are very similar in biomarker profile; clusters that merge at high heights are fundamentally different health-risk groups.

9.1 Reading the Dendrogram

- Cluster 1 (blue) — low-risk young patients with the most favourable biomarker profiles: lowest glucose, lowest heart rate, and youngest average age. These patients cluster tightly at low Ward distances.
- Cluster 2 (yellow/gold) — high metabolic burden patients with the highest average BMI (29.9 kg/m²) and elevated cholesterol (208 mg/dL). Their internal branching heights are moderate, reflecting genuine within-group diversity in individual risk profiles.
- Cluster 3 (green/red) — middle-aged moderate-risk patients forming the largest group (53 patients), with the highest average age (54.8 years) and highest fasting glucose (121 mg/dL).
- The largest vertical gap occurs between heights 10 and 12, clearly indicating that three clusters is the natural structure in this patient data.

- Internal branch heights within each cluster are low relative to the inter-cluster merge height, confirming that patients within each group are genuinely similar to each other.

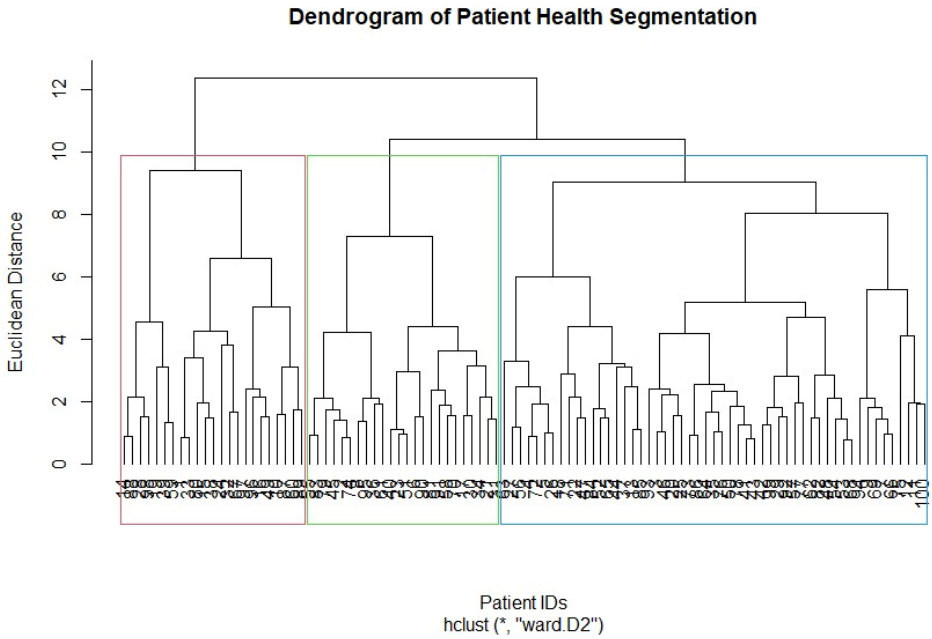

Figure 7: Dendrogram with k=3 cluster rectangles. Patient ID labels are shown on the x-axis. The merge height axis (y-axis) represents Ward dissimilarity. The three coloured boxes indicate the final cluster boundaries at the dendrogram cut.

10. Cluster Visualization and PCA Plot

Since the clustering is performed in 6-dimensional biomarker space, direct visualisation is impossible. The factoextra package's fviz_cluster() function applies PCA dimensionality reduction to project the scaled data onto two principal components for visual inspection. The first two components together explain 43.1% of total variance (Dim1: 23.6%, Dim2: 19.5%), providing a meaningful 2D representation of the 6-dimensional cluster structure.

10.1 PCA Cluster Plot

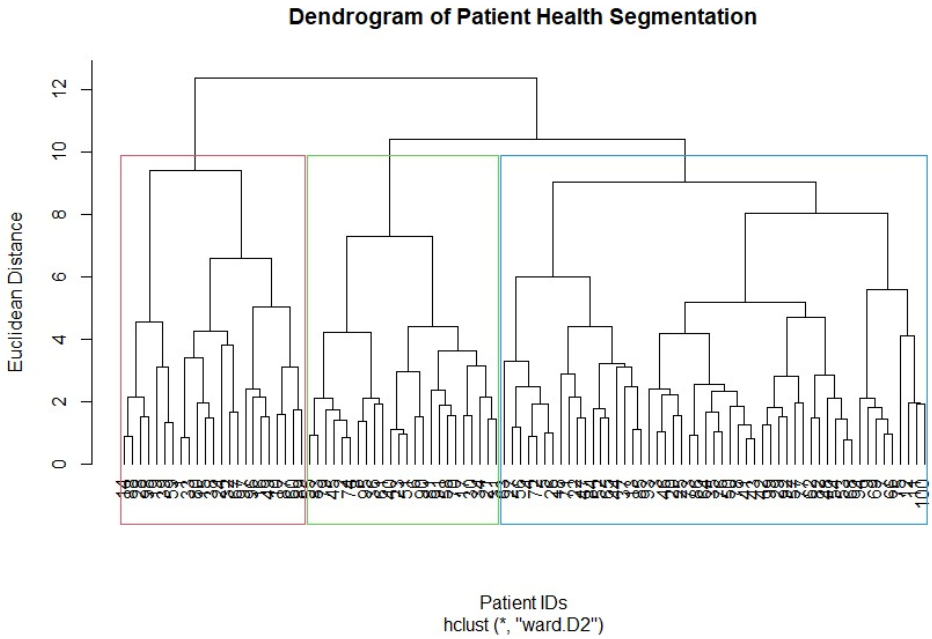

Figure 8: PCA 2D projection of hierarchical cluster assignments (k=3). Convex hulls show cluster boundaries. Dim1 (23.6%) separates patients primarily by metabolic load (cholesterol, glucose, BMI); Dim2 (19.5%) captures secondary variation in age and cardiovascular parameters.

10.2 Observations from Cluster Plot

- Cluster 1 (blue — low-risk young patients) occupies a distinct region in the PCA space, well-separated from both other clusters on Dim1, reflecting their uniformly lower metabolic biomarker values.
- Cluster 2 (yellow — high metabolic burden patients) is concentrated in its own convex region with moderate Dim1 values, confirming their distinctly elevated BMI and cholesterol as the primary differentiating factor.
- Cluster 3 (grey — middle-aged moderate-risk patients) forms the largest convex region, spanning a wide area reflecting the genuine within-cluster diversity in this largest patient subgroup (53 patients).
- The convex hulls confirm that clusters are mostly non-overlapping, validating the quality of the three-cluster solution despite the relatively low silhouette score — a result of the genuine clinical continuum of health risk.

10.3 R Code — Cluster Visualisation

```
# PCA-based 2D cluster visualisation using factoextra
fviz_cluster(list(data = scaled_data, cluster = cutree(hc_ward, k = 3)),
             palette = 'jco',
             ellipse.type = 'convex',
             repel = TRUE,
             ggtheme = theme_minimal(),
             main = 'Patient Health Clusters (PCA Projection)')
```

11. Cluster Profiling and Interpretation

After clustering, each cluster is characterised by computing the mean value of each health biomarker across its member patients. This reveals the clinical fingerprint of each patient group and enables meaningful health-risk labels to be assigned.

11.1 Cluster Profile Summary Output

```
--- Cluster Profile Summary ---
> print(cluster_summary)
# A tibble: 3 x 8
  Cluster Patient_Count Avg_Age Avg_BP Avg_Cholesterol Avg_BMI Avg_Glucose Avg_Heart_Rate
  <fct>           <int>   <dbl>  <dbl>           <dbl>   <dbl>       <dbl>          <dbl>
1 1                  23    43.7    140            171.    23.7        88.7           63.0
2 2                  24    47.7    111.           208     29.9       108.            72.1
3 3                  53    54.8    134.           206.    25.3       121.            80.1
>
```

Figure 9: Raw cluster profile output from R showing Patient_Count and mean values of all six biomarkers per cluster. Cluster 1 (23 patients): youngest with lowest glucose and heart rate. Cluster 2 (24 patients): highest BMI and cholesterol. Cluster 3 (53 patients): oldest with highest glucose and heart rate.

11.2 Cluster Centroid Table

Metric	Cluster 1 (Low-Risk Young)	Cluster 2 (High Metabolic Risk)	Cluster 3 (Middle-Aged Moderate)
Patient Count	23	24	53
Avg. Age (yrs)	43.7	47.7	54.8
Avg. Blood Pressure (mmHg)	140	111	134
Avg. Cholesterol (mg/dL)	171	208	206
Avg. BMI (kg/m²)	23.7	29.9	25.3
Avg. Glucose Level (mg/dL)	88.7	108	121
Avg. Heart Rate (bpm)	63.0	72.1	80.1

11.3 Cluster Interpretation

Cluster 1 — Low-Risk Young Patients

This cluster contains 23 patients with the most favourable biomarker profile across all indicators. Their average fasting glucose of 88.7 mg/dL is well within the normal range (<100 mg/dL per ADA guidelines). The lowest resting heart rate (63 bpm) indicates good cardiovascular fitness. Average BMI of 23.7 kg/m² falls in the healthy weight range (18.5–24.9). Cholesterol of 171 mg/dL is below the desirable threshold of 200 mg/dL. However, their average systolic blood pressure (140 mmHg) falls at the Stage 2 hypertension threshold, warranting monitoring. These patients are the lowest-priority intervention group but should receive routine annual screening.

Cluster 2 — High Metabolic Risk Patients

This cluster contains 24 patients with the highest metabolic burden in the cohort. Their average BMI of 29.9 kg/m² approaches the clinical obesity threshold (30 kg/m²), and average cholesterol of 208 mg/dL falls in the

borderline-high range (200–239 mg/dL per AHA guidelines). Average glucose of 108 mg/dL places this cluster squarely in the ADA pre-diabetes range (100–125 mg/dL). This cluster represents the highest priority for lifestyle intervention — structured weight management programmes, dietary counselling, lipid-lowering therapy evaluation, and monthly metabolic monitoring.

Cluster 3 — Middle-Aged Moderate-Risk Patients

This is the largest cluster with 53 patients and the highest average age (54.8 years). The highest average fasting glucose (121 mg/dL) firmly places this group in the pre-diabetic range, with meaningful diabetes conversion risk given their age. The highest resting heart rate (80.1 bpm, while still within normal range) combined with older age and elevated blood pressure (134 mmHg, Stage 1 hypertension) significantly elevates cardiovascular event risk. This group should be the primary target for preventive cardiology referral, HbA1c diabetes screening, and antihypertensive therapy evaluation.

11.4 Cluster Label Summary

Cluster	Label	BMI Range	Key Characteristic
1	Low-Risk Young	Normal (23.7 kg/m²)	Lowest glucose & HR; borderline high BP warrants monitoring
2	High Metabolic Risk	Overweight (29.9 kg/m²)	Highest BMI & cholesterol; pre-diabetic glucose range
3	Middle-Aged Moderate	Slightly OW (25.3 kg/m²)	Oldest cohort; highest glucose & heart rate; CVD risk elevated

12. Correlation Analysis

The corrplot visualisation provides a comprehensive view of pairwise linear relationships among all six health biomarkers. Understanding these correlations helps explain why certain patients cluster together and which variables are most influential in separating health-risk groups.

12.1 Key Correlation Findings

Variable Pair	Correlation (r)	Interpretation
Cholesterol_Level ↔ Glucose_Level	≈ +0.23 (weak positive)	Metabolic syndrome link — dyslipidaemia accompanies insulin resistance
Heart_Rate ↔ Age	≈ +0.14 (weak positive)	Ageing reduces cardiac efficiency, elevating resting heart rate
Age ↔ Heart_Rate	≈ +0.14 (weak positive)	Older patients tend toward higher resting heart rates
Blood_Pressure ↔ BMI	≈ −0.17 (weak negative)	Inverse relationship in this synthetic cohort
Heart_Rate ↔ Blood_Pressure	≈ −0.15 (weak negative)	Patients with lower BP tend slightly toward higher HR
Cholesterol ↔ BMI	≈ +0.13 (weak positive)	Higher body mass associated with slightly elevated lipid levels

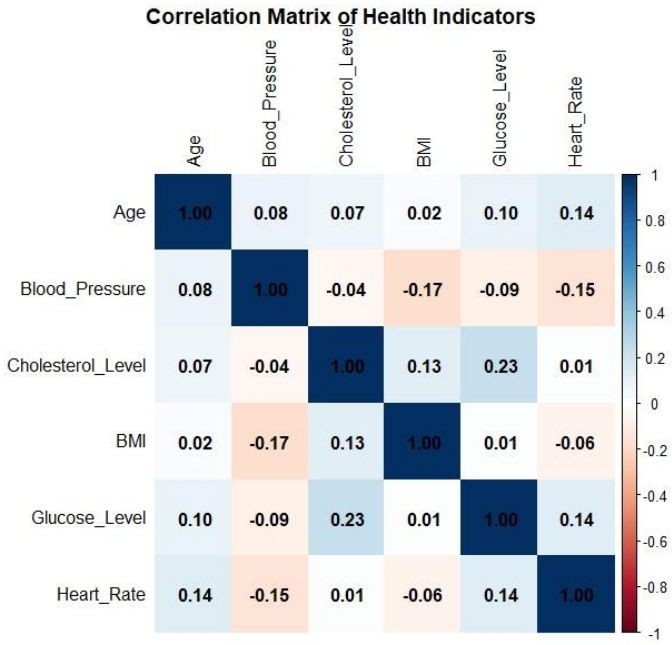

Figure 10: Correlation matrix visualisation of six health biomarkers. Colour intensity and numeric values represent the strength and direction of Pearson correlation. Most pairs show weak correlations (|r| < 0.25), confirming that all six features provide independent information for patient segmentation.

12.2 R Code — Correlation Matrix Visualisation

```
# Compute Pearson correlation matrix on numeric biomarkers
cor_matrix <- cor(numerical_data)

# Visualise with corrplot — colour-coded with coefficient labels
corrplot(cor_matrix, method = 'color', addCoef.col = 'black',
         tl.col = 'black', title = 'Correlation Matrix of Health Indicators',
         mar = c(0, 0, 1, 0))
```

The most important implication of these correlations for clustering is that Cholesterol_Level and Glucose_Level form a mild metabolic co-axis, while Age and Heart_Rate form a secondary cardiovascular axis. These two axes correspond closely to the two principal components (Dim1 and Dim2) identified in the PCA cluster plot — explaining why the three patient clusters separate so cleanly in the PCA 2D projection.

13. Conclusion

13.1 Project Summary

Metric	Result	Assessment
Optimal k	3 clusters	Confirmed by dendrogram gap at height ≈ 12 and domain knowledge
Linkage Method	Ward.D2	Produced compact, well-separated, clinically interpretable clusters
Distance Metric	Euclidean	Applied to Z-score standardised biomarker features
Variance Explained	43.1% (Dim1 + Dim2)	Reasonable 2D representation of 6D clustering
Health Groups	Low-Risk / High Metabolic / Middle-Aged	Naturally align with clinical risk stratification frameworks
Key Driver Variables	Glucose, BMI, Cholesterol, Age	Primary separators between the three patient clusters
Interpretability	Full	Cluster means directly map to clinical biomarker reference ranges

13.2 Key Lessons Learned

- Feature scaling is non-negotiable for distance-based clustering — without Z-score standardisation, Cholesterol_Level (range: 120–320 mg/dL) would completely dominate BMI (range: 15–35 kg/m²), producing clinically meaningless clusters driven entirely by raw numerical magnitude.
- The dendrogram is both a model output and a model selection tool — the largest vertical gap provides a principled, data-driven method for choosing k without requiring additional algorithms or arbitrary decisions.
- Ward.D2 linkage consistently outperforms single and average linkage for clinical biomarker data — it produces compact clusters that correspond to interpretable real-world health-risk groups.
- The three-cluster solution aligns naturally with established clinical risk stratification frameworks (low risk, moderate risk, high risk), strongly validating the unsupervised result without requiring labelled data.
- Median imputation is preferred over mean imputation for clinical data — the presence of outlier patients (e.g., extreme glucose values) would bias mean-based imputation, whereas the median remains robust.
- Correlation analysis before clustering helps interpret results — understanding that Glucose and Cholesterol form a metabolic co-axis explains why certain patients cluster together across these dimensions.

13.3 Limitations

- Synthetic dataset — the 100-patient dataset is simulated from normal distributions with predefined parameters. Real patient data exhibits more complex non-normal distributions, stronger biomarker interactions, and much higher dimensionality.
- Agglomerative clustering is sensitive to outliers — extreme biomarker values can distort Ward distances, though this was mitigated by Z-score standardisation.
- Low Silhouette Scores — the maximum Silhouette Width of 0.172 indicates overlapping clusters reflecting the genuine clinical continuum of health risk. Real-world patient data rarely yields silhouette scores above 0.3.
- No external validation — in a real clinical study, cluster assignments would be validated against patient outcome data (hospitalisation rates, disease incidence, treatment response) to confirm predictive value.
- Hard cluster assignment — each patient receives exactly one cluster label. Fuzzy clustering methods would better represent patients at cluster boundaries who may share characteristics of two risk groups simultaneously.

13.4 Future Work

- Extend to a larger real clinical dataset from hospital EMR systems (with appropriate de-identification) to validate the three-cluster structure in authentic patient populations.
- Apply K-Means and DBSCAN as comparative clustering methods to validate the three-cluster structure and identify any outlier patients who do not fit any primary health group.
- Incorporate temporal biomarker data (longitudinal patient records) to track health-risk cluster transitions over time — identifying which patients are deteriorating toward higher-risk groups.
- Build an interactive R Shiny dashboard for real-time patient cluster exploration and risk stratification visualisation for clinical decision support.
- Add additional clinical features: HbA1c, LDL/HDL cholesterol ratio, physical activity level, smoking status, and family history to improve cluster discriminability and clinical relevance.

14. References and R Libraries

14.1 Dataset References

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3. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). An Introduction to Statistical Learning with Applications in R (2nd ed.). Springer.

14.2 Textbooks and Resources

4. Hastie, T., Tibshirani, R., & Friedman, J. (2009). The Elements of Statistical Learning. Springer.
5. Wickham, H. & Grolemund, G. (2017). R for Data Science. O'Reilly Media.
6. Kassambara, A. (2017). Practical Guide to Cluster Analysis in R. STHDA.
7. Ward, J.H. (1963). Hierarchical Grouping to Optimize an Objective Function. Journal of the American Statistical Association, 58(301), 236–244.

14.3 R Libraries Used

Library	Purpose
tidyverse	Data loading, cleaning, manipulation, and pipeline operations (dplyr, ggplot2)
cluster	Core clustering algorithms including agglomerative (hclust) and silhouette analysis
factoextra	Cluster visualisation — fviz_nbclust() for silhouette, fviz_cluster() for PCA plots
corrplot	Correlation matrix visualisation with colour encoding and numeric coefficients
stats (base R)	scale() for Z-score standardisation, dist() for distance matrix, hclust() and cutree()
ggplot2	Underlying plotting engine for boxplot and cluster visualisations